

Sajad Abavisani

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Green Card pending — no sponsorship required

SUMMARY

Final-year PhD candidate in Electrical and Computer Engineering at Georgia Tech, with years of experience in machine learning, causal inference, and time series analysis. Eager to apply and further develop expertise in machine learning, deep learning, and advanced time series analysis to solve complex, real-world challenges.

EDUCATION

Ph.D. in Electrical & Computer Engineering

Aug 2020 – May 2026

Georgia Institute of Technology, Atlanta, GA — GPA 3.8/4.0

Focus: Causal Discovery, Time Series, Deep Learning, Optimization

Coursework: Mathematical Foundations of ML, ML with Limited Supervision, Conversational AI, Convex Optimization, Statistical ML, Information Theory, Random Processes

M.S. in Electrical & Computer Engineering

Aug 2020 – Aug 2023

Georgia Institute of Technology — GPA 3.9/4.0

B.S. in Electrical Engineering

Sep 2014 – Sep 2019

Ferdowsi University of Mashhad, Iran — GPA 4.0/4.0

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, R, Java, SQL, Clingo/ASP

ML & Deep Learning: PyTorch (CUDA), TensorFlow, Hugging Face (Transformers, PEFT, Datasets), LLM fine-tuning, RAG pipelines (LangChain, LlamaIndex), Graph Neural Networks (PyTorch Geometric), Transformers, GANs, Diffusion Models, Capsule Networks, Scikit-learn

Data & Infrastructure: Pandas, NumPy, SciPy, NetworkX, Statsmodels, Docker, AWS, Azure ML, W&B, MLflow, Git, GitHub Actions CI/CD, Linux, SLURM clusters

Specializations: Causal inference, constraint programming (ASP/Clingo), graph algorithms, time series analysis & forecasting, statistical modeling, Bayesian inference, optimization, open-source development (PyPI), agentic coding (Claude Code, Cursor)

RESEARCH EXPERIENCE

AI-Augmented Causal Discovery from Noisy Time Series | Georgia Tech

2024 – Present

- Engineered RnR, a meta-solver that refines outputs of any causal discovery algorithm by modeling undersampling effects via ASP, improving F1 scores by 45% over SOTA baselines (PCMCI, FASK, MVGC, GIMME). Under review at ICML 2026.
- Designed prioritized multi-stage optimization (density → latent structure → edge orientation), reducing solution space by ~4 orders of magnitude while maintaining global optimality guarantees. Validated on real fMRI data.
- Building an AI-enhanced classification pipeline (HC vs Schizophrenia) using Graph Transformers (BrainNetTransformer) and a novel Solution-Set Transformer that learns diagnostic biomarkers from RASL's multi-graph output — a capability unique to our method.
- Developing an LLM-guided framework to select ground-truth causal graphs from equivalence classes by combining chain-of-thought reasoning with domain knowledge, using fine-tuned models on simulated data with known ground truth.

Scalable Causal Discovery & Network Integration | Georgia Tech

2022 – 2024

- Extended causal discovery from 6-node toy graphs (prior SOTA) to structured graphs with 100+ nodes by exploiting SCC decomposition, achieving linear scaling in number of SCCs — enabling practical neuroimaging applications for the first time.
- Ran experiments on 19-machine SLURM cluster (64 cores, 512 GB RAM each) with Clingo in 10-thread parallel mode.
- Developed ION-C: an ASP algorithm for causal learning from overlapping datasets with non-co-measured variables. Proved soundness and completeness; scaled from 4–6 node (prior limit) to 25+ nodes. Validated on European Social Survey data.

Constraint-Based Causal Structure Learning (GRACE-C) | Georgia Tech

2020 – 2023

- Reformulated the rate-agnostic structure learning problem into a declarative ASP encoding, achieving 1,000× speedup (17 hrs → 6 sec). **ICLR 2023 Oral (top 25%)**.
- Proved correctness and completeness via direct encoding theorem; verified against 1,000 test graphs with identical outputs to provably correct RASL baseline.
- Built optimization mode with weighted constraint relaxation for noisy inputs; applied to real resting-state fMRI.
- Co-developed dRASL for deliberate undersampling: proved that measuring *more slowly* can reduce causal uncertainty by up to 4 orders of magnitude. Published at CLeaR/PMLR 2023.
- Released all methods as the pip-installable gunfolds open-source package. (pypi.org/project/gunfolds)

Deep Learning for Computer Vision & Medical Imaging | Georgia Tech / Rutgers

2019 – 2021

- Co-developed DECAPS for COVID-19 CT classification: 96.1% AUC with inverted dynamic routing and conditional GAN augmentation (pix2pix), outperforming three board-certified radiologists (95% vs 85% TPR at matched FPR).
- Co-authored Greedy AutoAugment: reduced augmentation policy search from exponential to linear complexity, 360× fewer GPU-hours than AutoAugment. Published in Pattern Recognition Letters.

Robotic Prosthetic Control | Ferdowsi University

2017 – 2019

- Designed multimodal neural networks (EMG + vision) for prosthetic gesture recognition using CNNs and restricted Boltzmann machines.

SELECTED INDUSTRY-RELEVANT PROJECTS

- **LLM-Guided Causal Graph Selection:** Building a framework using fine-tuned LLMs with chain-of-thought prompting to select ground-truth DAGs from equivalence classes, integrating domain knowledge with data-driven reasoning.
- **Solution-Set Transformer for Brain Classification:** Designing a two-level architecture (graph encoder + Set Transformer + classifier) that learns diagnostic biomarkers from multiple candidate causal graphs per subject — uniquely enabled by RASL's multi-solution output.
- **Net Worth & Portfolio Estimator:** Built a Python tool integrating Yahoo Finance API (yfinance), brokerage CSV import, and real estate valuation to compute portfolio value and net worth with per-position breakdown.
- **iOS Accent-Learning App:** Swift app leveraging OpenAI Whisper for speech recognition with quantitative pronunciation feedback.
- **LLM + Knowledge Graph Reasoning:** Integrated Llama 3 with knowledge graphs to improve logical reasoning on riddle-solving tasks.
- **GAN Robustness for Autonomous Driving:** Trained GANs to augment street-sign datasets, improving classification under adverse weather using multimodal inputs.

PUBLICATIONS

M. Abavisani, D. Danks, S. Plis. *GRACE-C: Generalized Rate Agnostic Causal Estimation via Constraints*. ICLR 2023 (**Oral, Top 25%**)

M. Abavisani et al. *RnR: A Meta-Solver for Causal Discovery in Undersampled Time Series*. Under review, ICML 2026.

K. Solovyeva, D. Danks, **M. Abavisani**, S. Plis. *Causal Learning through Deliberate Undersampling*. CLeaR / PMLR 2023.

P. Nair et al. (incl. **M. Abavisani**). *ION-C: Integration of Overlapping Networks via Constraints*. arXiv 2024.

A. Naghizadeh, **M. Abavisani**, D. Metaxas. *Greedy AutoAugment*. Pattern Recognition Letters 2020.

A. Mobiny et al. (incl. **M. Abavisani**). *Radiologist-Level COVID-19 Detection via Capsule Networks*. arXiv 2020.

HONORS

- ICLR 2023 Oral Presentation (Top 25% of accepted papers)
- Ranked 3rd in ACM Programming Competitions
- Top 0.001% (537th / 1,000,000+) in nationwide university entrance exams, Iran
- Accepted into NODET (National Organization for Development of Exceptional Talents)